

XRPL Validator — Daily Engineering Report • 2026-07-23

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **14 commits** (xrpl-validator 14)
- 1 test / vector commit

Engineering changes

- **fix(xrpl-ledger): enforce owner reserve on object creation — TicketCreate + TrustSet** (bcf7e5c , xrpl-validator) The native engine minted owned objects without checking the account could afford the incremental owner reserve, returning tesSUCCESS where mainnet returned tecINSUFFICIENT_RESERVE (TicketCreate) / tecINSUF_RESERVE_LINE (TrustSet). Closes ~22 fresh-traffic scout divergences in one class fix.
- **fix(xrpl-ledger): AMMDeposit owner-reserve guard for zero-LPToken depositors** (385e129 , xrpl-validator) Same owner-reserve class as the TicketCreate/TrustSet fix: an AMMDeposit by an account holding ZERO LPTokens opens (or funds) an LPToken trust line and so must keep XRP above accountReserve(ownerCount + 1). The native engine skipped this and over-applied (tesSUCCESS, 4-5 muts) where mainnet returned tecINSUF_RESERVE_LINE (net_muts=1).
- **fix(xrpl-ledger): direct IOU payment requires an existing destination trust line** (a6ef95a , xrpl-validator) A Payment delivers an IOU by rippling through the issuer, which credits the destination's EXISTING trust line — rippled never opens a trust line for the receiver. Our apply_iou_direct phantom-created the destination's line and delivered, returning tesSUCCESS (with extra RippleState + owner-dir + dir-page mutations) where mainnet returned tecPATH_DRY (fee-only), because the destination did not trust the issuer.
- **fix(xrpl-ledger): payments never autobridge — AMM over-mutation via a synthesized XRP leg** (4a36c48 , xrpl-validator) rippled synthesizes the XRP bridge only in FlowCross (offer crossing). A payment's strands come from its own Paths plus the default path, so it reaches a book only where its path names one. cross_engine_to bridged unconditionally for IOU<->IOU, inventing a route the transaction never asked for.
- **fix(xrpl-ledger): no AMM liquidity in permissioned-domain books — corpus 100%** (84ced29 , xrpl-validator) BookStep::tryAMM returns early when book_.domain is set ("amm doesn't support domain yet", BookStep.cpp:820), so a pool never participates in a domain-scoped book. cross_engine_to discovered the AMM regardless of domain, letting a domain offer cross a pool that rippled does not admit.
- **fix(xrpl-ledger): a payment cannot open the destination's trust line** (e27d088 , xrpl-validator) Every payment strand ends by rippling the delivered IOU from its issuer to the destination, and that final step is a DirectStepI whose check opens with "Since this is a payment a trust line must be present" — no line yields terNO_LINE (DirectStep.cpp:423). Because every strand shares that same last step, one missing line invalidates all of them, and a flow with no surviving strands is tecPATH_DRY.

- **fix(xrpl-ledger): charge the issuer's TransferRate on payment delivery** (2b1beae , xrpl-validator) The engine ignored TransferRate entirely, so every payment that moved an IOU between two non-issuers delivered the gross amount. rippled charges the issuer's rate on the strand's last step: DirectStepI issues to the destination and qualitiesSrcIssues sets srcQOut = transferRate(issuer) whenever the previous step redeems (DirectStep.cpp:765), which for a payment is always — BookStep reports DebtDirection::Redeems ...
- **fix(xrpl-ledger): AccountDelete fee is one owner reserve, read from the ledger** (3cebd6a , xrpl-validator) AccountDelete::calculateBaseFee returns calculateOwnerReserveFee(view, tx) — "the fee required for AccountDelete is one owner reserve". That comes from the ledger's fee settings and is not a constant: the 2024 vote cut the increment from 2 XRP to 0.2, so a hardcoded 2_000_000 rejects every present-day AccountDelete.
- **fix(xrpl-ledger): merging past an oracle's ten-entry cap is tecARRAY_TOO_LARGE** (bfc83f9 , xrpl-validator) OracleSet is an upsert. Overflowing kMaxOracleDataSeries by MERGING the transaction's series into the existing object is a tec, not a tem: the transaction is well formed and only the resulting object is too big, so the fee is claimed (OracleSet.cpp:168). temARRAY_TOO_LARGE is a separate preflight check on the transaction's own array (line 46), which is not what these hit.
- **fix(xrpl-ledger): correct the credential keylet, allow self-issuance, hydrate it** (08515b3 , xrpl-validator) Four faults stacked on one transaction, each hiding the next.
- **fix(xrpl-ledger): tecOBJECT_NOT_FOUND for missing NFT offers, tecAMM_BALANCE for over-withdraw** (ce27e48 , xrpl-validator) Two unrelated result-code gaps, both fee-only on mainnet.
- **feat(xrpl-ledger): DX_BOOK trace for the book walk's quality gate** (eb19a38 , xrpl-validator) Prints each directory's encoded quality against the taker's threshold at the point the walk decides to break, in the same style as DX_AMM and DX_BRIDGE. The crossing gate is a 64-bit encoded-rate comparison where a single ulp decides whether an offer is reachable, and there was no way to see either side of it without a rebuild.
- **fix(xrpl-ledger): reject offers by achieved quality and skip expired ones; keep the cleanup on tecKILLED** (a823d98 , xrpl-validator) Three faults in book crossing, uncovered together on two IoC offers that mainnet leaves completely untouched.

Tests and campaigns

- **test(xrpl-node): repeatable corpus sweep + restore the #105761560 fixture** (6b3b87b , xrpl-validator)

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Expiring Trust Lines with Reserve Sweep** (tokens) Add an optional sfExpiration (LedgerIndex) SField to RippleState, settable by either side via a new tfSetExpiration flag on TrustSet only when that si
- **Auditor-Keyed Disclosure for Confidential MPT** (privacy) Extend the confidential MPT transfer transaction (per XLS-96) with an optional sfAuditCiphertext blob (bounded to 512 bytes), populated only when the

- **Canonical Wrapped-Asset Object for XChainBridge** (interop) Add a new `ItWRAPPEDASSET` ledger object (keylet::wrappedAsset(bridgeID, issuer)) auto-created on a Bridge's first `XChainCommit`, tracking `sfOutstandingS`

Appendix — commits

Time	Repo	Commit	Subject
11:28	xrpl-validator	bcf7e5c	fix(xrpl-ledger): enforce owner reserve on object creation — TicketCreate + TrustSet
12:03	xrpl-validator	385e129	fix(xrpl-ledger): AMMDeposit owner-reserve guard for zero-LPToken depositors
16:37	xrpl-validator	a6ef95a	fix(xrpl-ledger): direct IOU payment requires an existing destination trust line
19:06	xrpl-validator	4a36c48	fix(xrpl-ledger): payments never autobridge — AMM over-mutation via a synthesized XRP leg
19:06	xrpl-validator	6b3b87b	test(xrpl-node): repeatable corpus sweep + restore the #105761560 fixture
19:37	xrpl-validator	84ced29	fix(xrpl-ledger): no AMM liquidity in permissioned-domain books — corpus 100%
20:20	xrpl-validator	e27d088	fix(xrpl-ledger): a payment cannot open the destination's trust line
21:29	xrpl-validator	2b1beae	fix(xrpl-ledger): charge the issuer's TransferRate on payment delivery
22:34	xrpl-validator	3cebd6a	fix(xrpl-ledger): AccountDelete fee is one owner reserve, read from the ledger
22:35	xrpl-validator	bfc83f9	fix(xrpl-ledger): merging past an oracle's ten-entry cap is tecARRAY_TOO_LARGE
22:35	xrpl-validator	08515b3	fix(xrpl-ledger): correct the credential keylet, allow self-issuance, hydrate it
23:05	xrpl-validator	ce27e48	fix(xrpl-ledger): tecOBJECT_NOT_FOUND for missing NFT offers, tecAMM_BALANCE for over-withdraw
23:10	xrpl-validator	eb19a38	feat(xrpl-ledger): DX_BOOK trace for the book walk's quality gate
23:54	xrpl-validator	a823d98	fix(xrpl-ledger): reject offers by achieved quality and skip expired ones; keep the cleanup on tecKILLED